

Environment

- Ubuntu 20.04
 - python:python 2.7
 - python3: python 3.8
- (more details please see Table 1 in the paper)

Project Locations

- Experiments with a natural noisy image: ./natural_experiment
- Experiments with simulated images of different noise strengths: ./forImages
- Experiments with the simulated noisy dataset: ./forDataset

Instructions

Experiments with a natural noisy image: you just need to run 1a_RunMe.sh

Here you will find the steps for the repetition of the quantitative image quality comparison:

But if you want to repeat the same results in the paper, please do not generate the simulated data again.

Experiment 1: Experiments with the simulated noisy dataset

(1 Noise Weights; 800 Images)

1. use Matlab or run them in the terminal by "matlab -nodesktop -nosplash -nojvm -r 'a_main.m;quit'" to run Matlab without a GUI.

a_main.m: (0)Data Generation (1)VSNR (2-5)LRR GSR UTV1 UTV2

a_average_SSIM.py: Testing the average SSIM of the Generation dataset

2. (6-14)for gwy methods:

python d_batch_align_rows.py

python d_batch_scar_remove.py

python d_batch_gwy2png.py

3. (15)DS2:

mkdir Sim_Noisy_Images_txt DS2 denoise_txt

python3 d_batFile_png2txt.py

cp ./destripe2_Linux64 ./Sim_Noisy_Images_txt/

cd ./Sim_Noisy_Images_txt

ulimit -s unlimited

sh bat.sh

find ./ -name 'DENOISE_*' | while read -r filename; do cp "\$filename" ../denoise_txt; done

cd ..

python3 d_bat_txt2png.py

4. (16)WNN: the results were generated by a GPU server

5. (17)Comparison Results Generation

python d_batch_PSNR_SSIM_values.py

python3 d_draw.py

Experiment 2: Experiments with simulated images of different noise strengths (files + their methods)

1. a_main.m: (0)Data Generation (1)VSNR (2-5)LRR GSR UTV1 UTV2 (use Matlab or run them in the terminal by "matlab -nodesktop -nosplash -nojvm -r 'a_main.m;quit'" to run Matlab without GUI)

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2. (6-14)gwy:

python d_batch_align_rows.py

python d_batch_scar_remove.py

python d_batch_gwy2png.py

3. (15)DS2:

python3 d_batFile_png2txt.py

cd Sim_Noisy_Images_txt

ulimit -s unlimited

sh bat.sh

cd ..

python3 d_bat_txt2png.py

4. (16)WNN: the results were generated by a GPU server

5. (17)Comparison Results Generation

python e_ssim.py

python3 e_curves.py